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Lipidomics and transcriptomics insight into impacts of microplastics exposure on hepatic lipid metabolism in mice

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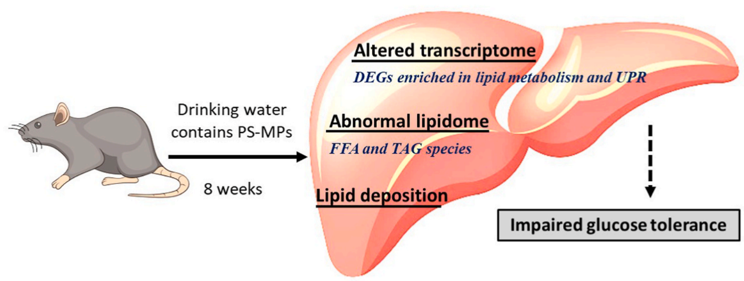
HIGHLIGHTS

- PS-MPs impaired glucose tolerance in mice.
- PS-MPs induced hepatic lipid deposition in mice.
- Hepatic lipid species, particularly FFAs and TAGs, were altered.
- DEGs were enriched in lipid metabolism process and unfolded protein response.

GRAPHICAL ABSTRACT



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ABSTRACT

Microplastics (MPs), the emerging environmental pollutants, have attracted global attention due to the potential public health challenge and ecological security risk. Recent studies suggested liver as a vulnerable organ to MPs exposure, evidenced by abnormal hepatic lipid metabolism upon MPs intake in multiple animal species. However, the specific changes of lipid metabolism in mammalian livers, as well as the underlying mechanisms, remain to be elucidated. In the present study, C57BL/6 mice were randomly assigned to normal drinking water or drinking water containing 100 μg L^{-1} or 1000 μg L^{-1} polystyrene (PS) MPs for 8 weeks. MPs exposure exerted no significant effect on body weight, serum triglyceride or total cholesteryl esters. However, mice showed impaired glucose tolerance and hepatic lipid deposition in response to high-dose MPs administration. Further lipidomic analysis showed significant alteration in hepatic lipid species particularly with free fatty acids (FFAs) and triacylglycerols (TAGs) in mice exposed to MPs. Meanwhile, the liver transcriptional profile indicated MPs exposure-induced differentially expressed genes (DEGs) were enriched in pathways of lipid metabolism and unfolded protein response. Furthermore, most altered lipid species were significantly correlated with DEGs enriched in lipid metabolic signaling. These findings provide lipidomic and transcriptional signatures of liver in response to MPs exposure, which will shed light on further understanding of the metabolic toxicity of MPs.

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1. Introduction

As an emerging environmental pollutant, microplastics (MPs) have attracted global attention in recent years. MPs are plastic fragments less than 5 mm in length, which are divided into primary or secondary MPs according to their sources (Vethaak and Legler, 2021; Wang et al., 2021; Bayar et al., 2022). Primary MPs come from industrial items such air fresheners, cosmetics, and personal health care products. Secondary MPs are fibers and fragments formed by biodegradation, mechanical abrasion, and photothermic destruction of bigger plastics in the environment (Wang et al., 2021). Main chemical compositions of MPs are various, including polystyrene (PS), polyethylene (PE), polyvinyl chloride (PVC), polypropylene (PP), and polyethylene terephthalate (PET) MPs, etc (Wang et al., 2021).

MPs were first discovered to be widespread in marine environment (Thompson et al., 2004) and could be enriched through the food chain of marine organisms (Cozar et al., 2014; Law and Thompson, 2014). Recent studies have shown the presence of large amounts of MPs in drinking water (Cox et al., 2019; Bauerlein et al., 2022), soil (Da Costa et al., 2019; Zhang et al., 2022), sediments (Irfan et al., 2020; Bayar et al., 2022), indoor/outdoor air (Prata, 2018; Batool and Qadir, 2022), and food products (seafood, sea salt, sugar, honey, etc.) (Liu et al., 2021). Furthermore, MP particles have been detected in human blood (Leslie et al., 2022), liver, kidney, lung (Amato-Lourenco et al., 2021), intestine, and even placenta (Ragusa et al., 2021). Ingestion through water and food is the predominant pathway for human MPs exposure (Prata et al., 2020). Epidemiological surveys have shown that the per capita intake of MPs is about 39,000–52,000 particles per year through digestive tract exposure (Cox et al., 2019). Therefore, MPs pollution is becoming a public health threat.

A variety of animal models, mainly aquatic animals, have been applied to explore the toxicity of MPs. The results show that the liver is a vulnerable organ to MPs exposure (Lu et al., 2016, 2018; Da Costa Araujo et al., 2020; Liu et al., 2022; Shen et al., 2022; Wang et al., 2022). Seven-day MPs treatment induced hepatic pathological damage, abnormal liver volume and cell morphology in tadpoles (Da Costa Araujo et al., 2020). Zebrafish exposed to MPs developed hepatic inflammation, oxidative stress, and lipid deposition (Lu et al., 2016). Recently, studies using mice and human liver organoids, also demonstrated abnormal liver lipid metabolism upon MPs challenge (Lu et al., 2018; Cheng et al., 2022). However, the specific changes and extent of lipid profiles in mammalian livers, as well as the underlying mechanisms, remain unknown.

PS-MPs are the predominant MPs type in the environment and are well applied in toxicological studies (Lu et al., 2018; Cheng et al., 2022; Liu et al., 2022; Wang et al., 2022). In the present study, we examined hepatic lipid profiles in mice exposed to PS-MPs to get a comprehensive understanding on the effects of MPs exposure on lipid metabolism. In addition, liver transcriptome was adopted to explore the potential mechanism of MPs-induced hepatic lipid metabolism disorders. We hope our study will shed light on the understanding of MPs metabolic toxicity in the liver.

2. Materials and methods

2.1. PS-MPs

According to recent reports about abnormal liver lipid metabolism caused by 1 μm PS-MPs (Cheng et al., 2022; Wang et al., 2022), we also used 1 μm PS-MPs in this study. Monodisperse PS-MPs, dispersed in deionized water with a concentration of 25 mg/mL, were acquired from Tianjin Baseline Chromatography Technology Development Center (Tianjin, China). PS-MPs stock solution was kept at 4 °C, and the microspheres were completely re-dispersed by sonication before use. The PS-MPs concentrations in the mouse treatment were 100 and 1000 μg L^{-1} according to previous studies (Lu et al., 2018; Zhao et al., 2022).

2.2. Animal MPs exposure

Male C57BL/6 mice (six-week-old) were acquired from Beijing Vital River Laboratory Animal Technology Co., Ltd (Beijing, China). All mice were kept in cages that were 22 ± 2 °C, had a 12-h light/dark cycle, and had free access to normal diet and water.

After acclimation, 24 mice were randomized into one of the three groups: control group (Con), low-dose group (Low), and high-dose group (High). The mice in low-dose and high-dose groups were treated with sterilized drinking water containing 100 μg L^{-1} or 1000 μg L^{-1} MPs, respectively, while the mice in control group were treated with the water alone. After eight weeks MPs exposure, mice were euthanized, body weight and liver weight weighted, and liver tissue fixed or frozen in liquid nitrogen and stored in an 80 °C freezer for further analysis. The Institutional Animal Care and Use Committee of Zhejiang Chinese Medical University gave its approval to the animal experimentation methodology.

During mouse exposure, we took several precautions to prevent potential MPs contamination. Specifically, avoid the use of plastic consumables when preparing normal and MPs-containing drinking water for mice. And all consumables were meticulously washed with distilled water and then sterilized to prevent airborne contamination. Besides, lab coats, sterile gloves, masks, and headgear were worn throughout the experiment.

2.3. Glucose tolerance test (GTT)

Mice were starved overnight after MPs exposure in order to measure blood glucose and perform an intraperitoneal glucose tolerance test (IPGTT). Blood glucose before and at 30, 60, 90, and 120 min after the dextrose (2 mg g^{-1} body weight) injection was detected with a FreeStyle Blood Glucose Meter (Abbott Diabetes Care Inc., Alameda, CA).

2.4. Serum triacylglycerols (TAGs) and total cholesteryl esters (TCEs) measurements

Whole blood was collected when the mice were euthanized. Serum was obtained via centrifugation at 2000 rpm for 10 min. TAGs and TCEs in serum were measured using a Hitachi 3100 Automatic Analyzer (Hitachi High-Tech Science Systems Inc., Tokyo, Japan).

2.5. Histological examination

Liver tissue was fixed in 4% paraformaldehyde (PFA) solution, paraffin-embedded, sectioned, regularly dewaxed, and stained with hematoxylin and eosin (H&E).

2.6. Oil red O staining

Fixed liver samples were embedded in an optimal cutting temperature (OCT) compound (Tissue-Tek, Sakura Finetek USA Inc., Torrance, CA, USA). The specimens were cut into thin slices (5 μm thickness) and stained with Oil red O for further lipid droplet detection. The percentage of Oil red O staining was measured by the ImageJ Program.

2.7. Lipid extraction

Briefly, 10 mg liver sample was homogenized with water (400 μL) and sonicated for 3 times. 10 μL of homogenate was combined with 190 μL water, then 480 μL of an extract solution containing an internal standard was added. The extract solution was the mixture of methyl-tert-butyl-ether and methanol (MTBE:MeOH, 5:1, v:v). The internal standards included 41 species from 12 classes, and the specific composition was shown in Table S1. 250 μL of supernatant was transferred to a fresh tube after vortex, sonication, and centrifugation. The remaining material was extracted twice with 250 μL of methyl *tert*-butyl ether. The mixed supernatants were then evaporated and reconstituted in 200 μL of resuspension buffer. After another round of vortex and sonication, the constitution was centrifuged at 12,000 rpm for 15 min at 4 °C. The final step was to transfer 40 μL of supernatant to a fresh glass vial for LC/MS analysis. An equal amount of the supernatants from each sample was combined to create the quality control (QC) sample.

2.8. Lipidomics analysis

All analytes were separated using a SCIEX ExionLC series UHPLC system (SCIEX, Framingham, MA, USA) with a Waters Acquity UPLC®HSS T3 column (1.8 μm, 2.1 × 100 mm). The mobile phase A was composed of 40% water, 60% acetonitrile, and 10 mM ammonium formate, while the mobile phase B was composed of 10% acetonitrile, 90% isopropanol, and 10 mM ammonium formate. The mobile phase parameters, including solvent gradient and run time, were list in Table S2. For assay development, an AB Sciex QTrap 6500+ mass spectrometer (SCIEX) was used. Molecules were ionized with electrospray ionization (ESI) source in positive/negative ion-switching mode. Typical ion source parameters were: ionspray voltage, +5500/-4500 V; curtain gas, 40 psi; temperature, 350 °C; ion source gas 1, 50 psi; ion source gas 2, 50 psi. For the quantification of the target chemicals, Biobud-v1.32 Software was applied. Based on the peak area and the actual concentration of the internal standard for the same lipid class, the absolute content of each lipid matching to the internal standard was estimated.

2.9. RNA isolation, library construction, and sequencing

Hepatic RNA was extracted with Trizol regent and qualified with Bioanalyzer 2100 system (Agilent Technologies, CA, USA). PolyA-tailed mRNA was enriched by Oligo (dT) magnetic beads, and then fragmented by fragmentation buffer. Fragmented mRNA was further reverse transcribed to synthesize cDNA. After end-repairing, A-tailing, and sequencing adapters connecting, 370–420 bp cDNA was screened with AMPure XP beads. Final library was obtained after PCR amplification and purification. Following qualification, different libraries were pooled, and then sequenced by the Illumina NovaSeq 6000 (Illumina, San Diego, CA, USA).

2.10. Sequencing data analysis

Image data measured by a high-throughput sequencer was converted to raw data by CASAVA base recognition. By eliminating adapter- and N-base-containing reads as well as low-quality reads from the raw data, clean data were obtained. The data were then mapped to the reference mouse genome using HISAT2 (v2.0.5). Differential expression analysis between groups was performed in DESeq2 R package (1.20.0). P-adjust ≤ 0.05 was set as the threshold for differentially expressed genes (DEGs). Gene ontology (GO) enrichment analysis of DEGs and statistical enrichment of DEGs in the Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway were achieved by clusterProfiler R package (3.8.1).

2.11. Transcriptome data validation

Six DEGs were randomly selected to validate the reliability of RNA-seq data. RNAiso Plus (Takara Biotechnology, Dalian, Liaoning, China) was used to isolate liver total RNA, and a cDNA Reverse Transcription Kit was used to produce cDNA (Takara Biotechnology). The QuantStudio Q7 Real-Time PCR machine was used to execute and quantify PCR reactions (Applied Biosystems, Carlsbad, CA, USA). Relative gene expression levels were calculated using a $2^{-\Delta\Delta CT}$ method. Primer sequences were listed in Table S3.

2.12. Statistical analysis

Experimental data are presented as mean ± standard error of the mean (SEM). Depending on the format of the data, the Student's t-test, or One-Way ANOVA followed by Dunnett's test, was employed to compare the exposed mice to the controls. Spearman correlation analysis was applied to evaluate the relationships between lipid profiles and transcriptional profiles. $P < 0.05$ was considered to be statistically significant.

3. Results

3.1. MPs exposure impaired glucose tolerance in mice

Body weight, serum lipids, fasting blood glucose, and glucose tolerance in mice were measured after MPs exposure. No significant difference in body weight, serum TAGs and TCEs levels was observed among those groups (Fig. 1A-1C). The mice in low-dose group showed similar fasting blood glucose and response to dextrose injection in GTT with the mice in control group (Fig. 1D-F). Interestingly, compared with the mice in control group, mice in high-dose group showed higher blood glucose 30 min and 60 min after dextrose injection and increased glucose AUC of GTT, albeit a significant reduction in fasting blood glucose (Fig. 1D and E). These observations suggested that MPs exposure impaired glucose tolerance in mice.

3.2. MPs exposure caused hepatic lipid deposition in mice

The coefficient of liver was not affected by eight weeks of MPs exposure (Fig. 2A). And no obvious difference was found in the hepatic morphology via H&E staining (Fig. 2B). However, compared with the mice in control group and low-dose group, mice administrated with high dose MPs showed accumulated intracytoplasmic lipids (Fig. 2C and D). These results indicated MPs exposure induced hepatic lipid deposition in mice.

3.3. MPs exposure disturbed hepatic lipid metabolites in mice

To get a comprehensive understanding on hepatic lipid alteration after MPs exposure, quantitative lipidomic analysis was conducted with total lipid extracted from liver in control and MPs-exposed mice. A total of 629 lipid species from 11 lipid classes of 5 categories, including 34 free fatty acids (FFAs), 333 TAGs, 45 diacylglycerols (DAGs), 13 cholesteryl esters (CEs), 15 ceramides (Cers), 15 hexosylceramides (HexCers), 12 sphingomyelins (SMs), 60 phosphatidylcholines (PCs), 74 phosphatidylethanolamines (PEs), 14 lyso-phosphatidylcholines (LPCs), and 14 lyso-phosphatidylethanolamines (LPEs), were identified in mouse liver.

No alteration of total lipid content was found in mouse liver after MPs exposure (Fig. 3A). It was worth noting that MPs treatment significantly increased the level of hepatic total FFAs (Fig. 3B). Compared with the control group, the level of hepatic FFAs in the low-dose and high-dose groups was increased by 20.2% and 26.4%, respectively. Further analysis of FFA subclass showed that the increase was observed in saturated fatty acids (SFAs), whereas the total amount of monounsaturated or polyunsaturated fatty acids (MUFAs or PUFAs) were not significantly altered with MPs exposure (Fig. 3C). At FFA specie level, low-dose and high-dose MPs exposures caused significant increase in 5 and 19 FFA species in the liver, respectively (Fig. 3D and Table S4 and S5).

Besides, a significant elevation of hepatic total PC content was observed in mice exposed to low-dose MPs while the content of hepatic PCs in the high-dose mice did not changed (Fig. 3E). No statistically significant difference was found in hepatic total DAGs, TAGs, CEs, Cers, HexCers, SMs, PEs, LPCs, and LPEs among different groups (Fig. 3E). Nevertheless, hepatic contents of some metabolites were elevated at the

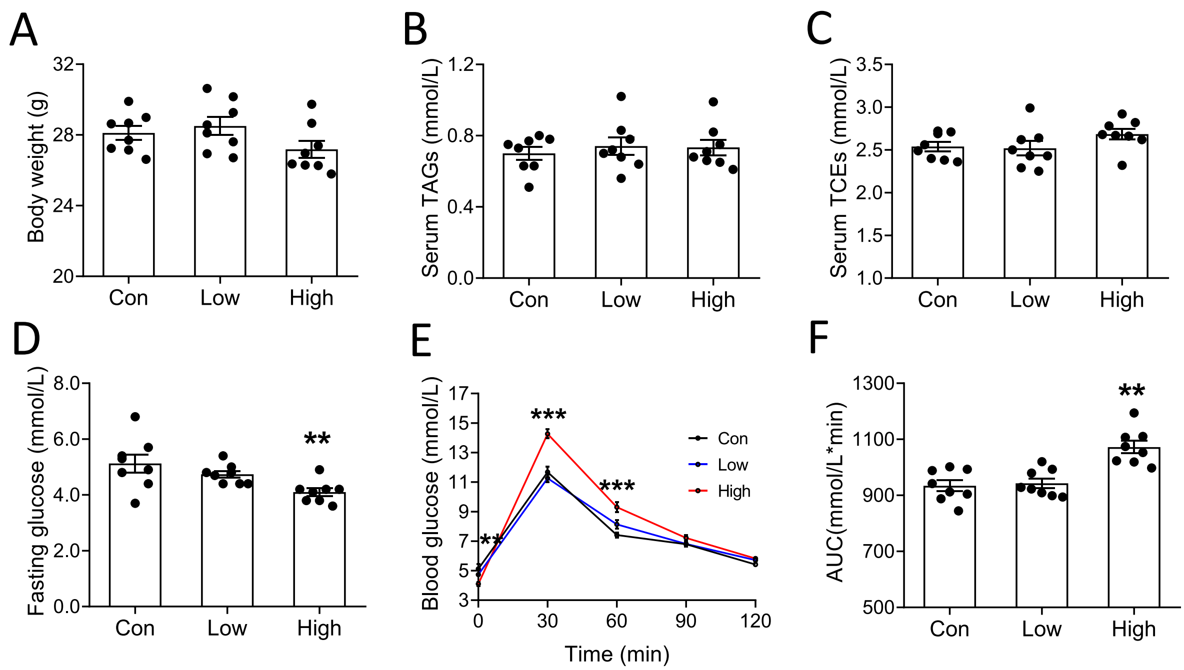

Fig. 1. Effects of MPs exposure on circulating lipid profiles and glucose homeostasis in mice. A. Body weight. B. Serum TAGs levels. C. Serum TCEs levels. D. Fasting blood glucose. E. Blood glucose in GTT in mice. F. GTT analysis with AUC. n = 8, $^{**}P < 0.01$, $^{***}P < 0.001$ compared with control group.

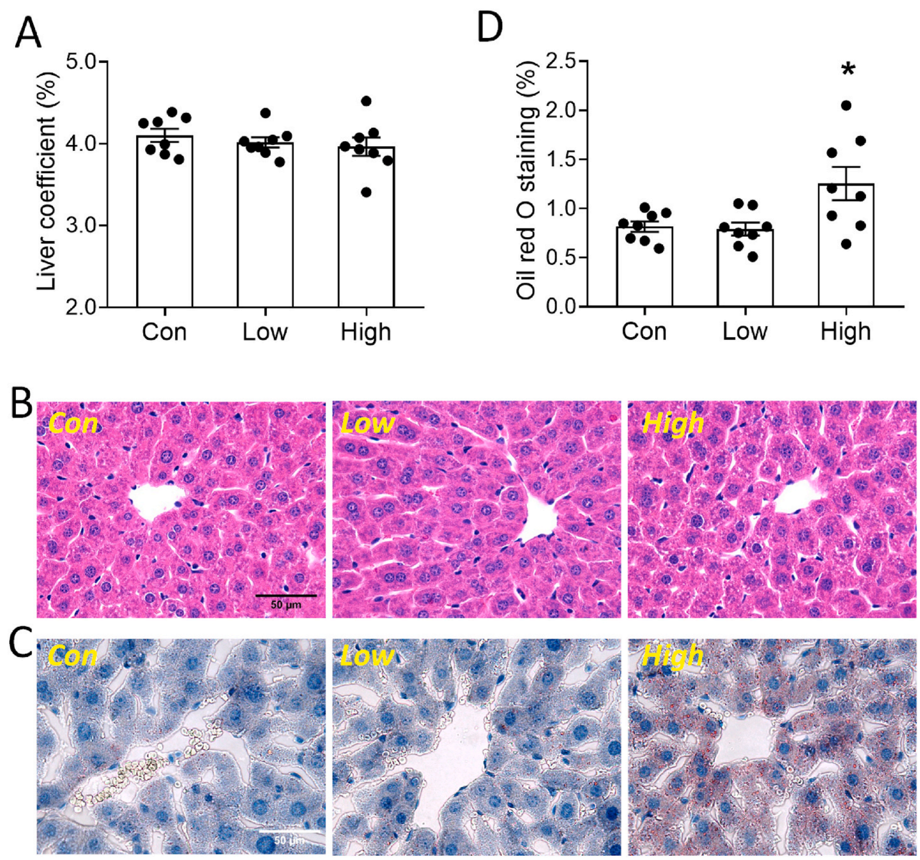

Fig. 2. Effects of MPs exposure on liver coefficient, morphology and lipid droplet in the liver of mice. A. Liver coefficient. B. Representative images of H&E-stained liver sections. C. Representative images of oil red O stained liver sections. D. Oil red O staining rate. n = 8, $^{*}P < 0.05$ compared with control group. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

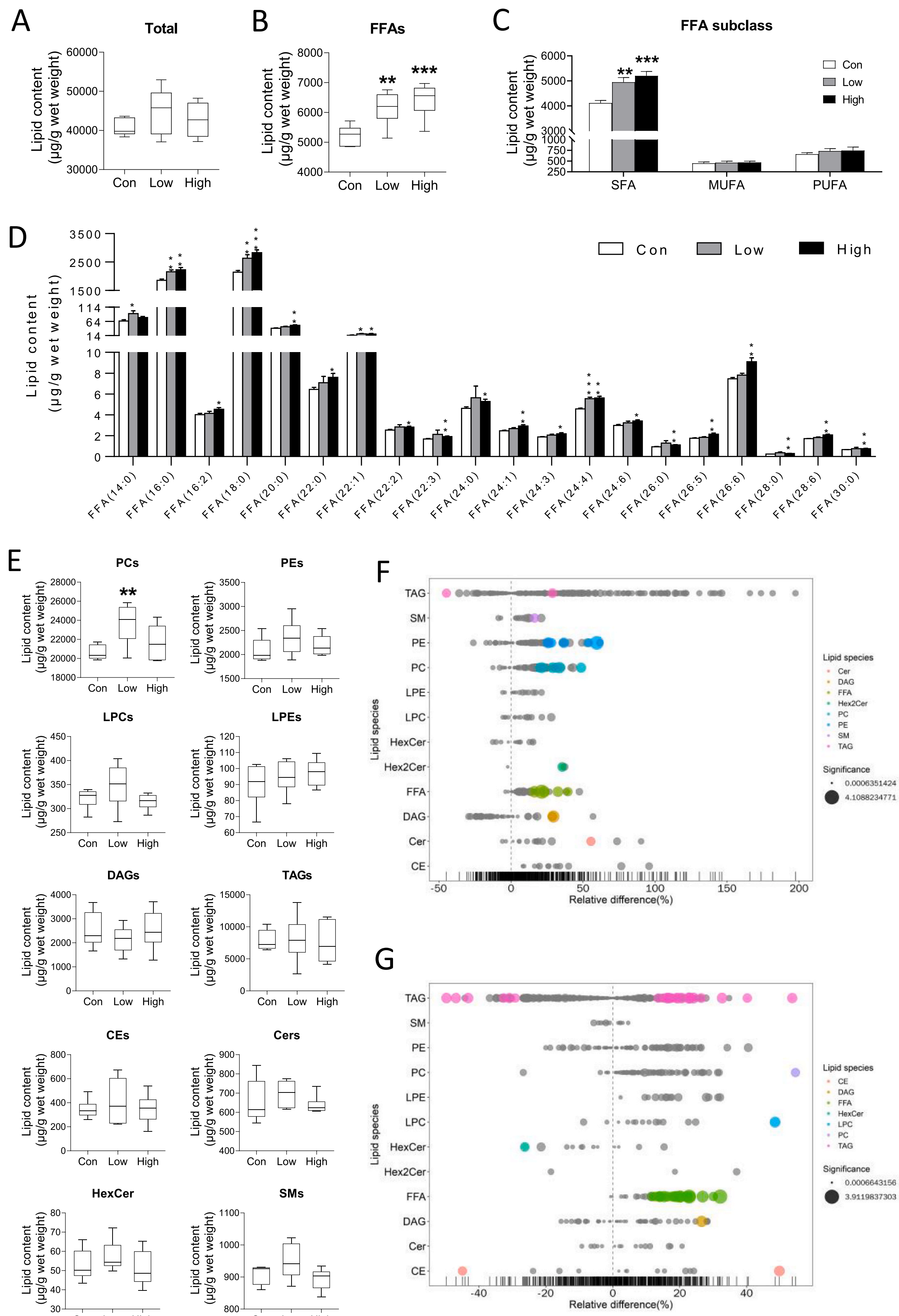

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Fig. 3. Effects of MPs exposure on hepatic lipid metabolites in mice. A-C. Total lipids (A), FFAs (B), and FFA subclasses (C) in the liver. D. Different FFA species in the liver. E. The contents of PCs, PEs, LPCs, LPEs, DAGs, TAGs, CEs, Cers, HexCers, and SMs in the liver. F and G. Bubble chart of hepatic altered lipid species in low-dose (F) and high-dose (G) groups. Each dot in the bubble plot represents a lipid species. The "significance" is the negative logarithm to base 10 of *P* values. The size of the dot represents the "significance", and the larger the dot, the smaller the value of the P-value. The gray dots represent non-significant differences with a *P*-value $>$ 0.05, and the colored dots represent significant differences with a *P*-value less than 0.05 (marked with different colors according to lipid classification). n = 6, $^{*}P <$ 0.05, $^{**}P < 0.01$, $^{***}P < 0.001$, compared with control group. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

specie level in mice exposed to low-dose MPs compared to control mice, including 6 PCs (PC of 18:0/18:0, 18:0/18:2, 18:0/18:3, 18:0/20:4, 18:0/22:5, and 18:0/22:6), 5 PEs (PE of P-16:0/18:2, P-16:0/20:4, P-18:0/18:2, P-18:0/22:5, and P-18:1/22:4), 2 DAGs (DAG of 14:0/20:0 and 16:0/18:0), TAG (50:5)_FA20:5, Cer (18:1/16:1), Hex2Cer (18:1/22:0), and SM (14:0). However, TAG (54:4)_FA22:4 level was reduced in the low-dose MPs group (Fig. 3F and Table S2). Moreover, mice administrated with high-dose MPs showed changes in 48 lipid species (Fig. 3G and Table S3), including 23 TAGs, 2 CEs, DAG (16:0/18:0), HexCer (18:0/22:1), PC (14:0/20:3), LPC (22:5), and 19 FFAs mentioned above. More specifically, 17 TAGs were evaluated after high-dose MPs exposure (Fig. 3G and Table S3). Taken together, these results demonstrated that MPs exposure could increase hepatic FFAs and some neutral lipids, glycerophospholipids, and sphingolipids species in mice.

3.4. Effects of MPs exposure on liver transcriptome

To explore the alterations in gene expression profile of mouse livers after exposure to MPs, transcriptome sequencing and analysis were performed. Compared to the control mice, 69 hepatic genes were significantly altered in the mice in the low-dose group, including 30 up-regulated genes and 39 down-regulated genes (Fig. 4A, Table S6); while 178 genes were significantly changed in mice exposed to high-dose MPs, of which 72 genes were up-regulated and 106 genes were down-regulated (Fig. 4B, Table S7). Moreover, when compared to the control group, 7 genes were up-regulated in both low- and high-dose groups (Figs. 4C), and 23 genes were down-regulated in both low-dose and high-dose groups (Fig. 4D). The heat map clustered 30 genes that were altered in the livers of mice in both high- and low-dose groups (Fig. 4E).

3.5. GO and KEGG analyses of DEGs

GO enrichment analysis was applied for functional classification of DEGs. In the comparison between low-dose group and control group, DEGs were significantly distributed in 346 GO items, including 257 BP (biological process) items, 38 CC (cellular component) items, and 51 MF (molecular function) items. And the top 20 GO items were mainly about mitochondrial electron transport chain, lipid metabolism and unfolded protein response (UPR) (Fig. 5A). In the comparison between the high-dose group and control group, 178 DEGs were distributed in 267 BP items, 50 C C items, and 72 MF items. Similar with the low-dose group, the top 20 GO items identified in the high-dose group were concentrated on lipid metabolism and UPR (Fig. 5B). Then KEGG pathway annotation analysis was conducted to further understand altered biological pathways caused by MPs exposure. DEGs between low-dose group and control group were enriched in 19 pathways (Fig. 5C), while 18 pathways were enriched by DEGs between the high-dose group and control group (Fig. 5D). The 10 common enrichment pathways of liver DEGs under different dose of MPs treatment were protein processing in fatty

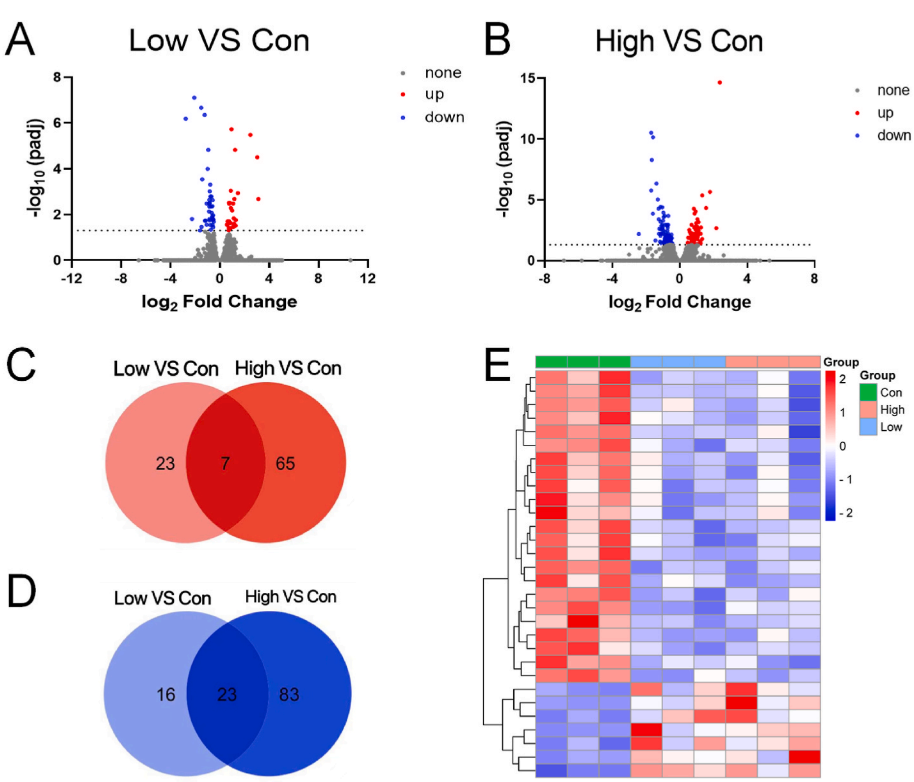

Fig. 4. Effects of MPs exposure on hepatic gene expression profile in mice. A and B. Volcano plot showing the number of up- and down-regulated genes in the low-dose group (A) and high-dose group (B) compared to the control group. C and D. Venn diagram of up-regulated genes (C) and down-regulated genes (D) in low-dose and high-dose groups. E. Heat map of genes that were altered in both high- and low-dose groups.

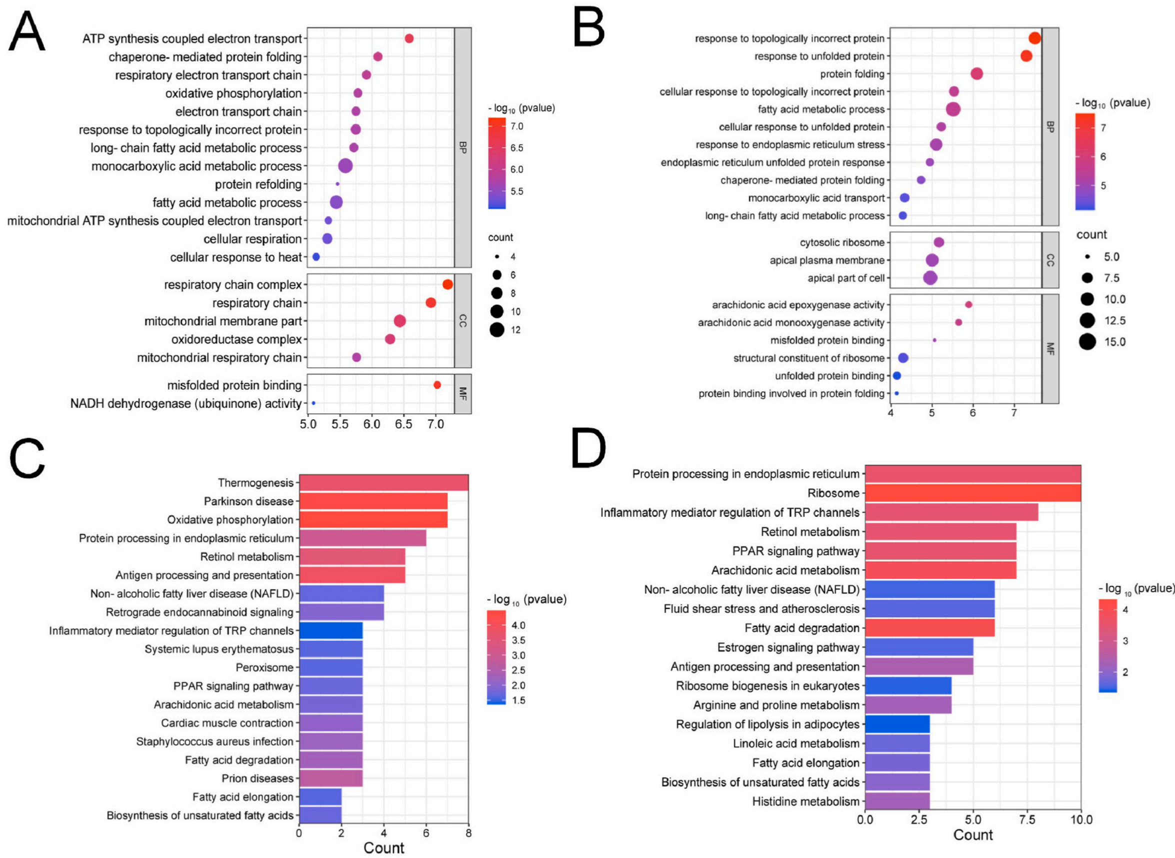

Fig. 5. GO classification and KEGG pathway annotation of DEGs. A and B. Top 20 GO items for DEGs between low-dose group (A) or high-dose group (B) and control group. C and D. Significant KEGG pathways for DEGs between low-dose group (C) or high-dose group (D) and control group.

acid degradation, arachidonic acid metabolism, biosynthesis of unsaturated fatty acids, fatty acid elongation, non-alcoholic fatty liver disease (NAFLD), PPAR signaling pathway, retinol metabolism, protein processing in endoplasmic reticulum, inflammatory mediator regulation of TRP channels, and antigen processing and presentation (see Fig. 5C and D).

3.6. transcriptome data validation results

Six DEGs, including 3 up-regulated genes and 3 down-regulated genes that were altered in both low-dose and high-dose groups, were randomly selected for qRT-PCR validation to confirm the reliability of transcriptome sequencing. In general, the expression patterns of these genes were consistent with the Illumina sequencing data, demonstrating that DEG analysis is credible (see Fig. 6).

3.7. Correlations between lipid profiles and transcriptional profiles

To investigate the correlation between lipid metabolites and transcript profiles, and reveal the genes involved in abnormal lipids metabolism in mouse liver upon MPs exposure, Spearman correlation analysis was performed for each altered lipid specie and DEGs. DEGs enriched in lipid metabolic process in GO analysis, including 10 genes in low-dose group vs control group and 16 genes in high-dose group vs control group, were used for correlation analysis.

In the analysis of the low-dose group and the control group, the results showed that the content of FFA species was negatively correlated with the levels of *Acot3*, *Acot4*, *Acsl1*, *Crat*, *Cyp2c67*, *Cyp2c68*, *Cyp2c69*, and *Hacl1*; the content of phospholipid species was negatively correlated with *Acot3*, *Acsl1*, *Cyp2c68*, and *Cyp2c69* levels; SM (14:00) negatively correlated with *Acot3*, Acot4, and *Acsl1* levels; and the only reduced TAG, TAG (54:4)_FA22:4 was positively correlated with *Crat*, *Cyp2c67*, *Cyp2c68*, *Cyp2c69*, and *Hacl1* levels (Fig. 7A). In addition, the expression levels of *Irs2* were positively correlated with most lipid species.

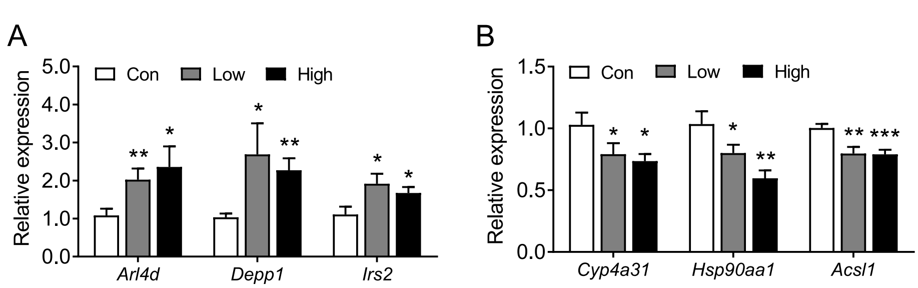

Fig. 6. Validation of DEGs expression in liver from mice exposed to MPs. A. Up-regulated genes were confirmed in both low-dose and high-dose groups. B. Down-regulated genes were confirmed in both low-dose and high-dose groups. n = 8, *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$, compared with control.

In the analysis of the high-dose group and the control group, the correlation between liver lipid species and altered genes was more extensive and complex (Fig. 7B). We found that the content of FFA species was negatively correlated with *Abhd2*, *Acot1*, *Acot3*, *Acot4*, *Acsl1*, *Cyp2c40*, *Cyp2c68*, *Cyp2e1*, and *Cyp4a10*. Similar inverse associations were observed for elevated TAG species and the above-mentioned genes. Besides, the content of elevated TAG species was positively correlated with the levels of *Etfb*, *Fabp1*, *Fads3*, *Irs2*, and *Ppargc1b*.

4. Discussion

The abnormal hepatic lipid metabolism induced by MPs exposure involves a complex toxicological process. In the present study, we evaluated the alteration of lipid metabolism and its pathways in mouse liver in response to two doses of MPs with high-throughput quantitative lipidomic and transcriptomic approaches. Our major findings include: (1) MPs exposure impaired glucose tolerance in mice; (2) MPs exposure induced hepatic lipid deposition and significant changes in lipid metabolites particularly with FFAs and TAGs in mouse liver; (3) MPs exposure changed hepatic transcriptional profile, with DEGs enriched in pathways of lipid metabolism and UPR.

MPs are ubiquitous in different environments and organisms, and pose potential hazards to various organisms, including humans (Vethaak and Legler, 2021; Wang et al., 2021). Previous studies have confirmed that liver is one of the main target organs of MPs toxicity (Lu et al., 2016, 2018). Orally exposed smaller MPs could cross the intestinal epithelium and translocate into the liver via the circulatory system (Meng et al., 2022). Existing research has demonstrated that the hepatotoxicity of MPs is primarily characterized by liver inflammation (Lu et al., 2016; Capo et al., 2021), oxidative stress (Capo et al., 2021; Fan et al., 2022), and metabolic abnormalities (Lu et al., 2018; Fan et al., 2022). In the present study, we found that 8-week MPs exposure yielded no alteration of liver weight or hepatocyte morphology in mice, whereas higher exposure dose caused mild but significant lipid deposition in the liver. Consistent with our findings, mice exposed to MPs for 20 weeks or even 180 days also had unchanged liver weight but increased oil red O staining (Fan et al., 2022; Shi et al., 2022a). However, Lu et al. discovered that mice exposed to MPs for 5 weeks had a decreased liver weight and a reduction in key liver lipids (Lu et al., 2018). These findings imply that the effects of MPs exposure on hepatic lipid deposition depend on the duration of exposure. Considering that human exposure to MPs is typically prolonged, MPs exposure may be a potential factor contributing to the disturbance of hepatic lipid metabolism to human.

Lipid metabolism, such as lipid acquisition, storage, and consumption, is one of the important functions of the liver. According to the LIPID MAPS classification system (Liebisch et al., 2020), fatty acyls (including FFAs and fatty acyl coenzymes As), glycerolipids, glycerophospholipids, sphingolipids, and sterol lipids are the main lipid categories in the liver. Although previous studies have reported the alterations of MPs exposure on hepatic TAGs and TCEs (Lu et al., 2018), no studies have investigated the effects on other lipid classes or specific lipid species in mouse liver. Using quantitative lipidomics, 11 lipid classes were identified and measured from 5 categories. Although MPs exposure did not induce significant increase in total lipid content in the liver, we found significant enhancement of hepatic FFAs levels in response to different dose of MPs exposures. It is worth noting that the increase was in SFAs subclass, while total amount of MUFAs and PUFAs was not affected by MPs exposure. Many previous studies have shown that appropriate amount of SFAs was required to maintain liver structure and lipid metabolism, but excess SFAs might disrupt homeostasis of endoplasmic reticulum (ER) and trigger UPR and ER stress (Wang et al., 2006; Cao et al., 2012). This was also confirmed by our subsequent transcriptome results, which demonstrated DEGs were mainly enriched in pathways or processes related to fatty acid metabolism and ER stress. Besides, a recent study also found that excess hepatic SFAs content was closely related to *de novo* hepatic lipogenesis and insulin resistance (Roumans et al., 2020). Therefore, the increase in SFAs found in our study may partly explain MPs induced hepatic lipid deposition and insulin resistance reported by other groups (Fan et al., 2022).

At FFA specie level, low-dose and high-dose MPs exposures induced significant increases in 5 and 19 FFA species in the livers of mice, respectively, which suggested a dose-response relationship in the increase of hepatic FFAs in response to MPs exposure. Considering the results of transcriptome and lipidome correlation analysis, we speculate

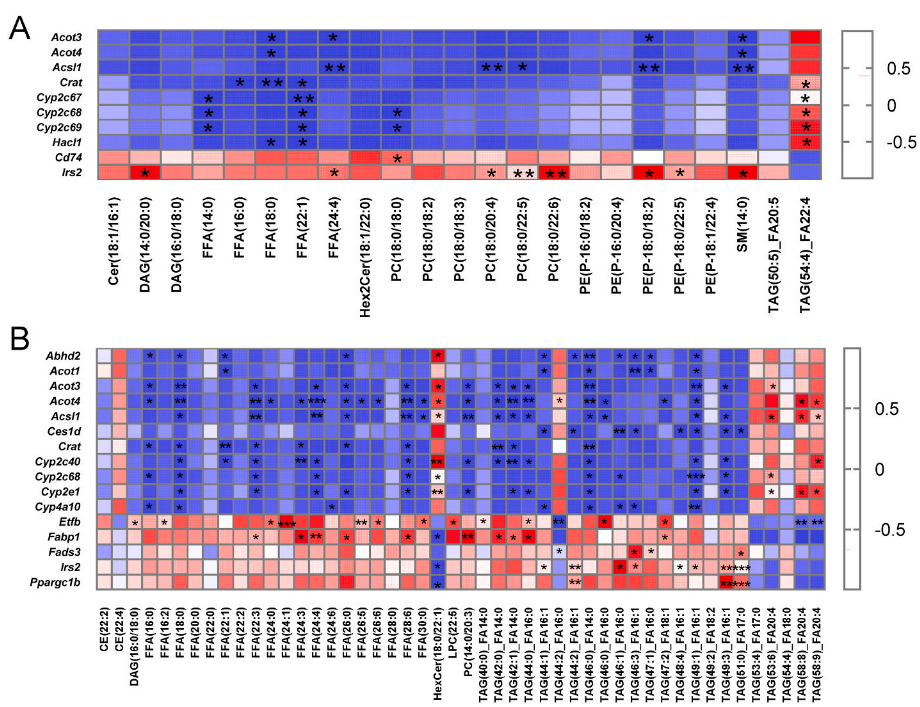

Fig. 7. Correlations between lipid profiles and transcriptional profiles. A. Correlation analysis when low-dose group was compared with the control group. B. Correlation analysis when high-dose group was compared with control group. $^{*}P < 0.05$, $^{**}P < 0.01$, and $^{***}P < 0.001$.

that this may be due to down-regulation of the expression of metabolic enzymes related to fatty acid transportation and oxidation (*Crat* and *Acsl1*), and elongation and decomposition (*Acot3* and *Acot4*) by exposure to MPs.

Using an off-target lipidome, Menendez-Pedriza et al. detected increased levels of 20 TAG species in MPs treated human hepatoma cells (HepG2) (Menendez-Pedriza et al., 2022). Similarly, our study found that TAG species in livers of mice were also significantly altered after exposure to MPs. And a dose-response relationship for the effects of MPs exposure on hepatic TAG species was observed. Exposure at the lower dose yielded changes in only 2 TAG species, while 23 TAG species were altered at the higher dose. In our present study, both low- and high-dose MPs treated mice displayed markedly elevated levels of TAG species with relatively lower carbon numbers and double-bond content, whereas reduced levels of TAG species with relatively high carbon numbers and double-bond content in the liver. The elevation of certain TAG species might be the result of down-regulated expression of lipid hydrolase such as *Abhd2* and *Ces1d,* and up regulation of *Irs2* in mouse liver upon MPs treatment. And the increased FFA species provided material basis for the synthesis of TAG. However, probably due to the down regulation of genes related to fatty acid elongation, the levels of TAG species with a high number of carbon atoms and double bonds were reduced. Many studies have shown that the risk of certain metabolic diseases was related to the number of carbon atoms and double bonds in TAG species (Rhee et al., 2011; Kihara, 2012; Stegemann et al., 2014). For example, the development of type II diabetes was inversely associated with TAG species with the number of carbon atoms and double bonds (Kihara, 2012; Stegemann et al., 2014). The lower number of carbon atoms and double-bond content in TAG species in the present study suggested that exposure to MPs might increase the risk of developing type II diabetes. Actually, we did observe impaired glucose tolerance, a precursor to type II diabetes, in mice exposed to high-dose of MPs. This is consistent with the study by Fan and colleagues (Fan et al., 2022), who found that MPs exposure caused both impaired glucose tolerance and insulin resistance in mice.

Previous transcriptome analysis showed that the expression of more than 8000 genes in *Gobiocypris rarus* liver was significantly altered after 28 days of MPs exposure (Wang et al., 2022), while 20 weeks of MPs ingestion upregulated 293 genes and downregulated of 351 genes in mouse liver (Fan et al., 2022). Possibly due to the short exposure time, less hepatic DEGs were observed in the present study. Nonetheless, the number of DEGs in the liver was increased as the exposure dose was increased. And the DEGs were closely linked to the lipid changes we observed. Besides, we found that some DEGs were enriched in UPR-related processes in ER, suggesting that ER stress may be a potential mechanism of MPs liver toxicity, which was worthy of further exploration.

Although our results provide lipidomic and transcriptional signatures of liver in response to MPs exposure, several questions remain. This study focused on the dose-dependent effects of MPs on hepatic lipid profiles and transcriptomes, but did not address the size effects of MPs on hepatic lipid metabolism. In published MPs exposure studies in mice, the size of MPs used ranged from tens of nanometers to tens of micrometers (Lu et al., 2018; Fan et al., 2022; Shi et al., 2022a, 2022b), whereas particles of 1 μm were used in this study. Although MPs with small particle sizes, particularly nano-sized MPs, are more likely to be deposited in the liver (Fan et al., 2022), it remains to be confirmed whether small-sized MPs can induce more severe liver lipid profile disorders and transcriptional abnormalities. Another issue is that MPs are ubiquitous in the environment and humans are at risk of long-term and even lifetime exposure. In this work, mice were only exposed to MPs for 8 weeks, and the reported alterations in liver lipids may not fully mirror the changes in human hepatic lipid metabolism following long-term MP exposure. In other published studies, mice were exposed to MPs for a maximum of 180 days (Shi et al., 2022a), which is far less than the human lifetime exposure. Therefore, the effects of chronic exposure to MPs, particularly exposure over a lifetime, on hepatic lipid metabolism remain to be revealed.

5. Conclusion

In conclusion, hepatic lipid metabolites and transcriptome were investigated in mice exposed to MPs by drinking water. The present study demonstrated that exposure to MPs disrupted the hepatic lipidome, manifesting primarily as elevated levels of FFA and TAG species. In the meantime, high-throughput transcriptome analysis revealed that the altered hepatic transcriptional profile was predominantly enriched in lipid metabolism and UPR-related pathways. These findings reveal the lipidomic and transcriptional responses of the liver after MPs exposure and shed light on the metabolic toxicity and potential mechanisms of MPs in the liver.

Author contributions statement

Qian Wang: Investigation, Methodology, Writing-Original Draft and Funding acquisition. **Yunlu Wu:** Investigation, Methodology and Writing-Original Draft. **Wenjing Zhang:** Investigation. **Ting Shen:** Investigation. **Haizhu Li:** Investigation. **Jingwei Wu:** Investigation. **Lu Zhang:** Methodology. **Li Qin:** Methodology. **Rucheng Chen:** Methodology. **Weijia Gu:** Methodology. **Mianhua Zhong:** Methodology. **Qinghua Sun:** Writing-Review & Editing. **Cuiqing Liu:** Conceptualiazation, Writing-Review & Editing and Funding acquisition. **Ran Li:** Conceptualiazation, Supervision, Writing-Review & Editing and Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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